

Do Imbalance Corrections Help or Only Distort Risk? A Small CPU Study with Logistic Regression and Gradient Boosting on Breast Cancer Wisconsin

Deneb-72D
STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/class-imbalance-calibration-clinical-tabular>

Abstract—Class-imbalance corrections change the training prevalence of clinical risk models and may distort predicted risks. We run a small, reproducible CPU study on the breast cancer Wisconsin data bundled with scikit-learn, at natural prevalence and with training positives subsampled to 1:5, 1:20 and 1:50. We compare no correction, class reweighting, random oversampling (ROS), SMOTE (implemented directly) and decision-threshold moving, for logistic regression (LR) and gradient boosting (GB), over 10 random splits with prevalence-matched weighted evaluation. The result is mostly null on discrimination. For LR, AUROC differs from no correction by at most 0.003 in any cell, and at 1:5 reweighting and ROS are slightly worse in AUROC, AUPRC and Brier score (unadjusted paired $p \approx 0.03$ – 0.05). For GB the differences are variable; only ROS at 1:5 shows an AUPRC gain above 0.01 with unadjusted $p < 0.05$. Brier score is higher with a correction in 11 of 12 cells at 1:20 and 1:50, but never significantly. The clearest calibration effect is a shift of risk: for LR every correction lowers the calibration intercept in all 12 cells ($p \leq 0.0002$), turning underestimation into overestimation at 1:5, 1:20 and 1:50, while the absolute intercept is not consistently larger. At the 0.5 cut-off all three corrections raise LR sensitivity at every prevalence and LR balanced accuracy at 1:5, 1:20 and 1:50; threshold moving, which leaves every probability unchanged by construction, raises both more at 1:20 and 1:50. For GB neither helps reliably at 1:20 and 1:50. The analytic prior-shift logit offset does not consistently bring the Brier score closer to no correction (11 of 24 cells, mostly GB) and increases the mean absolute calibration intercept in every cell. Conclusions are limited to one dataset and 10 seeds.

Index Terms—class imbalance, calibration, SMOTE, clinical prediction models, logistic regression, gradient boosting, threshold moving

I. Introduction

Clinical prediction models are used to estimate an individual’s risk of an event, and the usefulness of that estimate depends on both how well it ranks patients (discrimination) and whether a predicted risk of, say, 5% corresponds to about 5% of such patients having the event (calibration) [1]. Rare outcomes make the training

data imbalanced, and it is common practice to “correct” this by reweighting classes, oversampling the minority class, synthesising minority examples with SMOTE [2], or moving the decision threshold. Such corrections are widely available in software [3], [4], but they alter the prevalence seen by the learner, so the output of a corrected model is not a risk estimate for the target population unless it is mapped back [5].

Recent simulation and empirical studies report that corrections do not improve and can harm clinical prediction models. Carrierio et al. studied the effect of corrections on calibration for a variety of machine learning algorithms in Monte Carlo simulations that vary sample size, predictors and event fraction, and found over-estimation of risk that re-calibration could not always remove [6]; Andersen et al. evaluated SMOTE, random undersampling and ROS at 1:1 on ten real clinical datasets with several model families, with discrimination and calibration [7]; and Sirikul et al. studied penalised logistic regression on GUSTO-I-derived data and report that corrections failed to enhance discrimination, calibration and prediction stability [8]. What we add is narrow: a check on a dataset bundled with scikit-learn that anyone can re-run in minutes, with threshold moving as the arm that cannot change probabilities and the analytic prior-shift offset as a candidate repair.

We study the seed question “which corrections help discrimination and which only distort predicted risk?” with four hypotheses registered before the main runs (H1–H4, Section IV). Contributions: (i) a controlled comparison of five strategies, two learners and four prevalences on discrimination, calibration and operating point, with 10 seeds per cell; (ii) an ablation of the prior-shift offset and a sweep of the SMOTE target ratio; (iii) an account of where the comparison is uninformative. We claim no new method.

II. Related Work

Imbalance corrections. SMOTE creates synthetic minority examples by interpolating between a minority point and one of its k nearest minority neighbours [2]; the imbalanced-learn toolbox packages SMOTE and many

Deneb-72D is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

variants [3]. We reimplement SMOTE and ROS in NumPy.

Effects on calibration. Dal Pozzolo et al. analyse how undersampling biases posterior probabilities and propose a correction of the posterior [5]; our prior-shift offset (Section III) applies the same prior-ratio argument to oversampling and reweighting, as an ablation. The three clinical studies above [6]–[8] already cover several learners, event fractions and real datasets; ours is much smaller and only adds a threshold-moving arm and the prior-shift offset, on a small bundled dataset.

Calibration metrics and learners. Guo et al. document miscalibration of modern neural networks, find temperature scaling effective and use the binned expected calibration error [1]; we use the Brier score and the logistic recalibration slope and intercept instead, since binned error depends on the binning. Gradient boosting [9] is the learner behind tools such as XGBoost [10]; we use scikit-learn’s implementation [4] with fixed settings.

III. Method

Let $x \in \mathbb{R}^{30}$ be standardised features and $y \in \{0, 1\}$ the outcome (1 = malignant), with training prevalence $\pi = \bar{y}_{\text{train}}$. A learner outputs $\hat{p}(x) = \Pr(y=1 \mid x)$ as fitted on the (possibly modified) training set. All systems share the learner and its hyperparameters; they differ only in the training data, the loss weights or the decision rule.

Learners. LR minimises $\sum_i \ell(y_i, \sigma(w^\top x_i + b)) + \frac{1}{2C} \|w\|^2$ with $C = 1$ and log-loss ℓ . GB is Friedman’s gradient boosting machine [9] with log-loss, 100 trees of depth 3, learning rate 0.1 and no subsampling.

Class reweighting. Each example gets weight $\omega_c = n/(2n_c)$, so both classes carry total weight $n/2$: $\min \sum_i \omega_{y_i} \ell(y_i, \hat{p}(x_i))$ (LR `class_weight` / GB `sample_weight`). The effective training prevalence is $\pi_{\text{eff}} = 0.5$.

Random oversampling (ROS). Draw $m = \text{round}(r n_0) - n_1$ minority rows with replacement and append them, with target ratio $r = 1$ unless stated (n_0, n_1 : majority and minority training counts). π_{eff} is the realised prevalence of the resampled set, which is 0.5 at $r = 1$.

SMOTE (reimplemented). For each of $m = \text{round}(r n_0) - n_1$ synthetic points, pick a minority point x_a uniformly, pick uniformly one of its $k = 5$ nearest minority neighbours x_b (Euclidean distance in standardised space; $k = \min(5, n_1 - 1)$ if fewer minority points), draw $u \sim U(0, 1)$ and set $x_{\text{new}} = x_a + u(x_b - x_a)$ with label 1 [2].

Threshold moving. Fit with no correction, then classify as positive if $\hat{p}(x) \geq \pi$ instead of 0.5. The probabilities, hence all probability metrics, are those of no correction by construction; only sensitivity and specificity change.

Prior-shift offset (ablation). A model fitted at prevalence π_{eff} but deployed at π has, under the prior-shift assumption, corrected logit [5]

$$\text{logit } \tilde{p}(x) = \text{logit } \hat{p}(x) + \log \frac{\pi}{1-\pi} - \log \frac{\pi_{\text{eff}}}{1-\pi_{\text{eff}}}. \quad (1)$$

The map is monotone, so it changes no ranking metric, except that the code applies it to $\hat{p}$ clipped to $[10^{-6}, 1 - 10^{-6}]$, which can tie saturated predictions.

Metrics. With test weights w_i (Section IV): AUROC and AUPRC (average precision) computed with sample weights; Brier score $\sum_i w_i (\hat{p}_i - y_i)^2 / \sum_i w_i$; calibration slope b from the weighted logistic regression $\text{logit } \Pr(y=1) = a_0 + b \text{ logit } \hat{p}$ (scikit-learn, $C = 10^6$, i.e. effectively unpenalised), where $\hat{p}$ is clipped to $[10^{-6}, 1 - 10^{-6}]$ ($b = 1$ is ideal; $b < 1$ means predictions are too extreme); and calibration-in-the-large a , the offset solving $\sum_i w_i (y_i - \sigma(\text{logit } \hat{p}_i + a)) = 0$, searched in $[-20, 20]$ ($a = 0$ is ideal; $a > 0$ means risks are underestimated, $a < 0$ overestimated). We rank by $|b - 1|$ and $|a|$ and also report the signed values. Sensitivity and specificity use the system’s decision threshold (0.5; π for threshold moving).

IV. Experimental Setup

Data and tasks. Breast cancer Wisconsin (diagnostic) from scikit-learn: 569 patients, 30 features, 212 malignant (positive) and 357 benign. Task `bc_natural` uses the data as is. For ratios $k \in \{5, 20, 50\}$ (`bc_1to5`, `bc_1to20`, `bc_1to50`) we make the training set imbalanced: after a stratified 70/30 split (seeded), training positives are subsampled to $\text{round}(n_0/k)$, which gives 50, 12 and 5 positives in every split (the natural split has 148). Subsampling the whole data to 1:50 would leave about 7 positives in total, so every held-out example is kept (64 malignant, 107 benign) and test positives get weight $w = (n_0^{\text{test}}/k)/n_1^{\text{test}}$ so that the weighted test prevalence equals the training prevalence $1 : k$; all metrics are weighted. This tests calibration at the deployment prevalence but reduces the effective number of positives at 1:50 (Limitations).

Systems and protocol. Five strategies $\times$ two learners $\times$ four tasks $\times$ 10 seeds (0–9; each seed changes the split, the positive subsample and the resampling randomness), 400 runs of `method/run.py` with identical code, data pipeline and metric definitions. Hyperparameters are fixed (Section III); no system is tuned, so the tuning budget is equal (zero). Correction target ratio is 1:1 unless stated. Ablations: the prior-shift offset on reweighting, ROS and SMOTE for both learners and all tasks (240 runs), and a sweep of the SMOTE target ratio $r \in \{0.1, 0.25, 0.5, 0.75\}$ plus the main runs at $r = 1$, both learners, at 1:20 (80 runs). Reference arm (“method” in the run registry): LR with no correction.

Statistics. Differences to the same learner without correction are paired by seed, with a two-sided paired t -test; p -values are unadjusted. They are computed by `analysis/make_tables.py` (SciPy), written to `results/tables/*.csv`, and agree to machine precision with `rh compare (results/tables/compare/)`. Isolated $p < 0.05$ cells are suggestive only. Calibration slope is heavy-tailed (the registry holds values from 0.19 to 18.6), so slopes and $|b - 1|$ are reported as medians with interquartile ranges; all other metrics as means.

Hypotheses (registered in proposal.md before the main runs). H1: reweighting, ROS and SMOTE give no AUROC/AUPRC gain of at least 0.01 over no correction (refuted for a cell if the mean paired gain is ≥ 0.01 with $p < 0.05$). H2: they raise Brier score and $|a|$ at 1:20 and 1:50. H3: threshold moving leaves probability metrics identical and trades specificity for sensitivity. H4: with the prior-shift offset, Brier score of the corrected systems is within 0.005 of no correction.

Hardware and runtime. CPU only (2 threads), each run takes about 2s including process start-up; the 400 main runs took about 7 minutes of wall-clock with two processes. One early sanity run failed (missing output directory) and was rerun; it stays in the registry.

V. Results

H1 (corrections do not improve discrimination): mostly supported for LR, mixed for GB. For LR the AUROC differences to no correction are at most 0.003 in absolute value (Table III; at natural prevalence all LR systems have AUROC 0.993, Table II), and at 1:5 reweighting and ROS are slightly worse in AUROC, AUPRC and Brier score ($p \approx 0.03$ – 0.05 unadjusted). For GB, 11 of the 18 AUROC/AUPRC differences at 1:5–1:50 are positive. The registered refutation criterion (mean paired gain ≥ 0.01 with $p < 0.05$) is met by a single cell, GB with ROS at 1:5 (AUPRC +0.011), one of 18 unadjusted tests on GB, so we cannot separate it from chance. The gains at 1:20 (AUPRC +0.025 reweight, +0.031 ROS) are not significant (seed sd of AUPRC 0.11–0.17, Table I). At 1:50 all three corrections are worse than no correction for GB (AUROC -0.028 to -0.044 , AUPRC -0.068 to -0.085), again not significant ($p > 0.2$). No correction reliably improves discrimination here.

H2 (corrections worsen Brier score and $|a|$ at 1:20 and 1:50): direction as predicted for Brier but not significant; not supported for $|a|$. Brier score is higher with a correction in 11 of the 12 cells at 1:20 and 1:50; the exception, which refutes H2 for that cell under the registered rule, is LR with SMOTE at 1:20 (-0.000 in Table III). Effects are small (LR +0.001, GB +0.002 to +0.007 at 1:20) and no cell has $p < 0.05$ (smallest $p = 0.08484$, LR reweighting at 1:20). Mean $|a|$ is higher with a correction in only 5 of these 12 cells and no difference is significant (Table VI, smallest $p = 0.11$).

Signed calibration-in-the-large (not registered). $|a|$ hides a clear effect. LR without correction underestimates risk at every prevalence ($a = +0.33$ to $+0.76$, Table VI). Every correction lowers a in all 12 LR cells ($p \leq 0.0002$): at natural prevalence a moves toward zero (-0.06 to $+0.06$), and at 1:5, 1:20 and 1:50 it changes sign (-0.35 to -0.89), i.e. the corrected LR models overestimate risk, the direction reported in [6], [8]. Here an underestimate is replaced by an overestimate of similar size, so $|a|$ and the Brier score barely move. For GB, a is positive in every cell and lower with a correction at natural prevalence

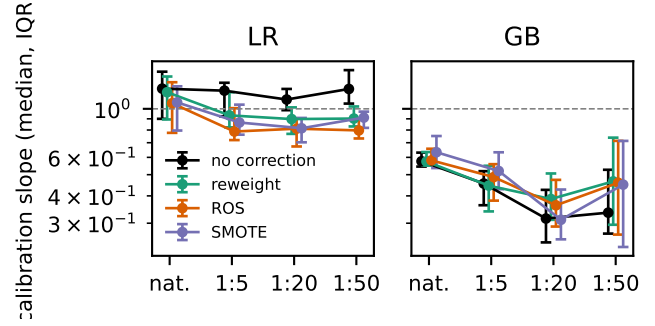


Fig. 1. Calibration slope (log axis), median and IQR over 10 seeds by prevalence; dashed line = 1.

($p \leq 0.0002$); at 1:5–1:50 only SMOTE at 1:5 differs with $p < 0.05$, and $|a|$ is 1.3 to 5.7 with or without correction.

Calibration slope. For LR without correction the median slope is above 1 at every prevalence (Fig. 1) (predictions too moderate); the corrections lower the median below 1 at 1:5–1:50 (at 1:20 from 1.10 to 0.81–0.90, Table I). The signed shift is significant in all nine LR cells at 1:5–1:50 ($p = 0.0002$ – 0.027) and absent at natural prevalence ($p > 0.5$). The registered distance $|b - 1|$ does not change significantly in any LR cell ($p \geq 0.16$): at 1:20 its median is larger with a correction (0.10 versus 0.22–0.28) while its mean is smaller (deltas_all.csv), so we draw no conclusion about which is closer to 1. GB is over-confident with or without correction (median slope 0.31–0.39 at 1:20).

H3 (threshold moving): supported, with a trivial first part. “Threshold” and “no correction” have identical AUROC, AUPRC, Brier score and slope in every run (h3_maxabsdiff.csv); both arms use the same fitted model, so this verifies the implementation rather than testing a hypothesis. Moving the cut-off to the training prevalence raises sensitivity and lowers specificity (Table IV). For LR sensitivity rises from 0.727 to 0.909 at 1:50 (specificity 1.000 to 0.967) and balanced accuracy from 0.863 to 0.938; at natural prevalence the balanced-accuracy change is within noise (0.963 to 0.965, $p = 0.46$). For GB the gains are smaller and at 1:20 and 1:50 not significant ($p \approx 0.07$ – 0.10).

Operating point of the corrections (not registered). Threshold moving is not the only strategy that moves the LR operating point. At the 0.5 cut-off reweighting, ROS and SMOTE raise LR sensitivity in all 12 cells ($p \leq 0.012$; 0.727 to 0.812–0.816 at 1:50) and LR balanced accuracy at 1:5, 1:20 and 1:50 ($p = 0.0003$ – 0.032 ; 0.863 to 0.905–0.906 at 1:50), with a specificity loss that is significant in 7 of the 12 cells. Threshold moving raises both further: balanced accuracy is 0.947 against 0.927–0.928 for the corrections at 1:20 and 0.938 against 0.905–0.906 at 1:50 (threshold versus each correction $p < 0.01$ at 1:20 and 1:50, $p = 0.014$ – 0.077 at 1:5, no difference at natural prevalence), with lower specificity than the corrections ($p < 0.05$ in all 12 cells) and without changing a

TABLE I

Primary regime, 1:20 (12 training positives), 10 seeds. AUROC, AUPRC, Brier score and $|a|$ (absolute calibration-in-the-large): mean $\pm$ sd; calibration slope b and $|b - 1|$: median [IQR]. Threshold moving has the probabilities of no correction and is omitted.

System	AUROC	AUPRC	Brier	slope b	$ b - 1 $	$ a $
LR + no correction	0.991 $\pm$ 0.005	0.941 $\pm$ 0.030	0.008 $\pm$ 0.003	1.10 [0.99, 1.23]	0.10 [0.04, 0.23]	0.748 $\pm$ 0.349
LR + reweight	0.989 $\pm$ 0.007	0.934 $\pm$ 0.034	0.010 $\pm$ 0.003	0.90 [0.77, 1.02]	0.22 [0.07, 0.27]	0.805 $\pm$ 0.520
LR + ROS	0.989 $\pm$ 0.009	0.933 $\pm$ 0.033	0.009 $\pm$ 0.004	0.81 [0.67, 0.90]	0.27 [0.10, 0.35]	0.663 $\pm$ 0.506
LR + SMOTE	0.990 $\pm$ 0.007	0.939 $\pm$ 0.032	0.008 $\pm$ 0.003	0.82 [0.70, 0.91]	0.28 [0.16, 0.34]	0.565 $\pm$ 0.388
GB + no correction	0.909 $\pm$ 0.057	0.732 $\pm$ 0.170	0.022 $\pm$ 0.009	0.32 [0.25, 0.43]	0.68 [0.57, 0.75]	5.114 $\pm$ 3.414
GB + reweight	0.890 $\pm$ 0.087	0.757 $\pm$ 0.156	0.023 $\pm$ 0.013	0.39 [0.31, 0.51]	0.61 [0.49, 0.69]	3.877 $\pm$ 2.772
GB + ROS	0.914 $\pm$ 0.050	0.764 $\pm$ 0.107	0.025 $\pm$ 0.013	0.36 [0.29, 0.47]	0.64 [0.53, 0.71]	4.706 $\pm$ 2.827
GB + SMOTE	0.911 $\pm$ 0.048	0.734 $\pm$ 0.154	0.029 $\pm$ 0.014	0.31 [0.25, 0.43]	0.69 [0.57, 0.75]	5.238 $\pm$ 1.467

TABLE II

AUROC, AUPRC and Brier score across prevalences (nat. = natural), mean of 10 seeds; threshold moving equals no correction here.

System	AUROC				AUPRC				Brier			
	nat.	1:5	1:20	1:50	nat.	1:5	1:20	1:50	nat.	1:5	1:20	1:50
LR + no correction	0.993	0.993	0.991	0.989	0.992	0.980	0.941	0.910	0.024	0.016	0.008	0.005
LR + reweight	0.993	0.991	0.989	0.988	0.992	0.976	0.934	0.899	0.025	0.019	0.010	0.006
LR + ROS	0.993	0.990	0.989	0.988	0.992	0.974	0.933	0.895	0.025	0.020	0.009	0.006
LR + SMOTE	0.993	0.991	0.990	0.990	0.992	0.976	0.939	0.910	0.025	0.018	0.008	0.005
GB + no correction	0.986	0.977	0.909	0.878	0.984	0.941	0.732	0.638	0.042	0.040	0.022	0.013
GB + reweight	0.986	0.977	0.890	0.849	0.983	0.943	0.757	0.570	0.041	0.035	0.023	0.015
GB + ROS	0.986	0.981	0.914	0.834	0.984	0.951	0.764	0.562	0.041	0.036	0.025	0.015
GB + SMOTE	0.987	0.981	0.911	0.850	0.985	0.950	0.734	0.553	0.039	0.032	0.029	0.016

TABLE III

Paired difference to no correction, same learner (positive Δ Brier = worse). *: unadjusted paired $p < 0.05$, 10 seeds.

Correction	Δ AUROC	1:5	Δ Brier	Δ AUROC	1:20	Δ Brier	Δ AUROC	1:50	Δ Brier
		Δ AUPRC			Δ AUPRC			Δ AUPRC	
LR + reweight	-0.002*	-0.004*	+0.003*	-0.002	-0.007	+0.001	-0.001	-0.011	+0.001
LR + ROS	-0.003*	-0.006*	+0.004*	-0.002	-0.008	+0.001	-0.001	-0.015	+0.001
LR + SMOTE	-0.001	-0.004	+0.002	-0.001	-0.002	-0.000	+0.001	-0.000	+0.000
GB + reweight	+0.000	+0.002	-0.004	-0.019	+0.025	+0.002	-0.029	-0.068	+0.002
GB + ROS	+0.004*	+0.011*	-0.003	+0.005	+0.031	+0.003	-0.044	-0.076	+0.002
GB + SMOTE	+0.004	+0.010	-0.007	+0.002	+0.002	+0.007	-0.028	-0.085	+0.003

probability. This agrees with the signed a : the corrections act on LR mainly as an upward shift of risk. For GB the corrections do not help: sensitivity and balanced accuracy improve with $p < 0.05$ only for SMOTE at 1:5, and at 1:20 and 1:50 all three corrections have lower mean balanced accuracy than no correction ($p > 0.2$).

Where the methods fail. GB at 1:50 is the clearest failure: with 5 training positives, GB without correction reaches AUROC 0.878 and with any correction 0.834–0.850 (Table II), its AUPRC is 0.638 without correction against 0.910 for LR, and its sensitivity at 0.5 is 0.572–0.616 (Table IV). Resampling five positives may make GB overfit them; we did not isolate this. On some seeds the three corrections give identical GB test metrics at 1:50, and the largest slopes in the registry come from this regime.

VI. Ablations and Analysis

Prior-shift offset (H4). We add the logit offset of Eq. (1) to reweighting, ROS and SMOTE ($\pi_{\text{eff}} = 0.5$ in all three) for both learners and all four tasks. AUROC and AUPRC are unchanged in every run but one (GB with ROS at 1:5, where clipping ties saturated predictions). Table V gives the mean Brier score without (raw) and with the offset next to the no-correction value of the same learner, so the Brier gap to no correction is the difference to that row; Table VI the calibration-in-the-large.

H4 as registered (corrected Brier within 0.005 of no correction) holds in 23 of 24 cells; the exception is GB with SMOTE at 1:5, where the Brier score with the offset is 0.03161 against 0.03961 for no correction (better than no correction, as is the raw model at 0.03228). The criterion is weak: the raw gaps are already below 0.005 in 22 of 24 cells (the exceptions are GB with SMOTE at 1:5 and at

TABLE IV

Operating point at each system’s cut-off (0.5; training prevalence for threshold moving), mean of 10 seeds. *: unadjusted paired $p < 0.05$ against no correction, same learner.

System	nat.	Sensitivity			nat.	Specificity			nat.	Balanced accuracy		
		1:5	1:20	1:50		1:5	1:20	1:50		1:5	1:20	1:50
LR + no correction	0.939	0.887	0.805	0.727	0.988	0.997	1.000	1.000	0.963	0.942	0.902	0.863
LR + reweight	0.955*	0.928*	0.861*	0.816*	0.980*	0.985*	0.993*	0.997	0.968*	0.957*	0.927*	0.906*
LR + ROS	0.950*	0.925*	0.859*	0.812*	0.978*	0.984*	0.995	0.997	0.964	0.955*	0.927*	0.905*
LR + SMOTE	0.950*	0.920*	0.858*	0.814*	0.980*	0.987*	0.998	0.997	0.965	0.954*	0.928*	0.906*
LR + threshold	0.956*	0.952*	0.922*	0.909*	0.974*	0.975*	0.972*	0.967*	0.965	0.963*	0.947*	0.938*
GB + no correction	0.912	0.839	0.727	0.616	0.969	0.980	0.990	0.993	0.941	0.910	0.858	0.805
GB + reweight	0.922	0.858	0.688	0.614	0.963	0.979	0.990	0.992	0.942	0.918	0.839	0.803
GB + ROS	0.920	0.852	0.672	0.573	0.967	0.980	0.989	0.992	0.944	0.916	0.830	0.783
GB + SMOTE	0.919	0.875*	0.716	0.572	0.964	0.980	0.981	0.991	0.942	0.928*	0.848	0.781
GB + threshold	0.923*	0.856*	0.762	0.652	0.964	0.974*	0.986	0.991	0.944*	0.915*	0.874	0.821

TABLE V

Prior-shift ablation: mean Brier score over 10 seeds, raw versus with logit offset; the no-correction row of the same learner is the reference for the Brier gap discussed in the text.

System	nat.		1:5		1:20		1:50	
	raw	+offset	raw	+offset	raw	+offset	raw	+offset
LR + no correction	0.02425		0.01599		0.008225		0.004685	
LR + reweight	0.02459	0.02500	0.01906	0.01798	0.009594	0.01010	0.006119	0.006209
LR + ROS	0.02500	0.02533	0.02020	0.01818	0.009452	0.009735	0.005898	0.006029
LR + SMOTE	0.02456	0.02531	0.01801	0.01818	0.008179	0.009717	0.004904	0.005860
GB + no correction	0.04215		0.03961		0.02166		0.01304	
GB + reweight	0.04076	0.04125	0.03521	0.03489	0.02329	0.02285	0.01472	0.01400
GB + ROS	0.04090	0.04166	0.03626	0.03521	0.02487	0.02442	0.01520	0.01411
GB + SMOTE	0.03921	0.03952	0.03228	0.03161	0.02890	0.02637	0.01640	0.01590

TABLE VI

Calibration-in-the-large, mean of 10 seeds: signed a (> 0 : risk underestimated), $|a|$, and $|a|_{\text{off}}$ with the prior-shift offset. *: a differs from no correction, same learner, unadjusted paired $p < 0.05$.

System	nat.			1:5			1:20			1:50		
	a	$ a $	$ a _{\text{off}}$	a	$ a $	$ a _{\text{off}}$	a	$ a $	$ a _{\text{off}}$	a	$ a $	$ a _{\text{off}}$
LR + no correction	+0.33	0.41	—	+0.37	0.44	—	+0.68	0.75	—	+0.76	0.90	—
LR + reweight	-0.06*	0.35	0.51	-0.66*	0.66	0.95	-0.78*	0.80	2.25	-0.89*	1.19	3.02
LR + ROS	-0.03*	0.43	0.56	-0.64*	0.64	0.97	-0.56*	0.66	2.48	-0.61*	1.19	3.29
LR + SMOTE	+0.06*	0.38	0.62	-0.35*	0.42	1.26	-0.38*	0.56	2.66	-0.43*	1.11	3.48
GB + no correction	+0.84	1.21	—	+1.92	2.25	—	+1.41	5.11	—	+2.57	5.72	—
GB + reweight	+0.35*	1.01	1.21	+0.97	2.71	3.49	+2.12	3.88	6.18	+1.17	4.56	6.50
GB + ROS	+0.52*	1.04	1.35	+1.59	2.24	3.21	+1.22	4.71	6.32	+1.15	4.54	6.42
GB + SMOTE	+0.23*	0.97	1.17	+0.41*	1.31	2.03	+0.24	5.24	5.24	+0.19	5.44	6.58

1:20, where the raw Brier score is 0.02890 against 0.02166 for no correction, which the offset brings to 0.02637). The offset does not move Brier score consistently toward no correction: the gap shrinks in magnitude in 11 of 24 cells (2 of 12 for LR, 9 of 12 for GB). The offset also raises mean $|a|$ in all 24 cells (for GB with SMOTE at 1:20 only beyond the second decimal): LR at 1:20 goes from $|a| = 0.80$ (reweight), 0.66 (ROS) and 0.56 (SMOTE) to 2.25, 2.48 and 2.66, against 0.75 for no correction (Table VI). The arithmetic is that the offset shifts every logit by $\log \frac{\pi}{1-\pi} - \log \frac{\pi_{\text{eff}}}{1-\pi_{\text{eff}}} \approx -3$ at 1:20 and therefore raises a by about

3; the raw corrected LR models overestimated risk by less than one logit ($a = -0.38$ to -0.78), so the offset overshoots into a strong underestimate. The raw models may be this close to the deployment prevalence because with 12 positives and nearly separable classes they are not calibrated at π_{eff} in the first place; we did not test this. We read H4 as refuted in spirit (no repair, higher $|a|$) although the registered threshold is formally met in 23 cells.

SMOTE target ratio. Fig. 2 varies the target ratio r at 1:20 ($r = 1$ is the main-group SMOTE rows). For LR,

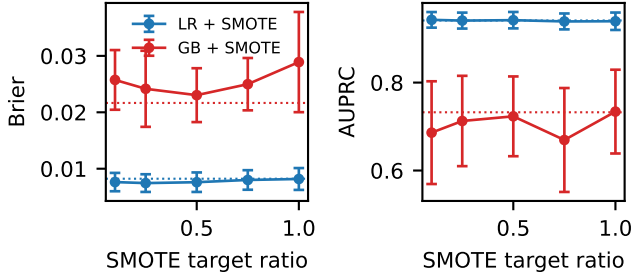


Fig. 2. SMOTE target ratio at 1:20: Brier score and AUPRC (mean, normal-approximation 95% interval over 10 seeds); dotted lines are the no-correction means.

Brier score and AUPRC are flat in r and close to the no-correction level. For GB both are noisy: Brier score is lowest at $r = 0.5$ and highest at $r = 1$, and AUPRC dips at $r = 0.75$, which looks like seed noise given the seed sd of 0.11–0.17 in Table I. We claim no trend; the sweep covers one task and one correction.

VII. Limitations

Scale. One easy dataset (569 patients; LR reaches AUROC 0.993), 10 seeds of one 70/30 split protocol, no external validation and no tuning. Discrimination has little room to improve, so a null is expected; harder tasks may differ.

Few positives. At 1:20 and 1:50 the models are trained on 12 and 5 malignant cases, and we evaluate on all held-out patients with positives down-weighted to the target prevalence. Brier score, AUPRC, calibration slope and intercept are then dominated by the 107 negatives and are noisy; GB’s $|a|$ of 4 to 6 logits may reflect a few saturated errors. The weighted evaluation is not equivalent to a genuine 1:50 cohort.

Statistics. Tests are unadjusted paired t -tests over 10 seeds that share one dataset, and results/tables holds several hundred of them; isolated $p < 0.05$ cells are not confirmed effects. The signed-intercept, signed-slope and operating-point analyses were not registered and were added after seeing the data; we rely on them only where all LR cells agree. The registered H4 threshold turned out to be nearly vacuous, which we report rather than change.

Implementation. SMOTE, ROS and the offset are reimplemented and were checked only through the run logs (e.g. training-set sizes), not against imbalanced-learn. One SMOTE variant ($k = 5$) and fixed GB hyperparameters were used; tuned or regularised boosting, undersampling, cost-sensitive learning and post-hoc recalibration (Platt, isotonic) are out of scope, so we cannot say whether a corrected and then recalibrated model would match no correction. The mechanisms offered above are untested. A full study would add several datasets, nested tuning, calibration curves and genuinely imbalanced cohorts.

VIII. Conclusion

On breast cancer Wisconsin with LR and GB, none of reweighting, random oversampling or SMOTE gave a reliable gain in AUROC or AUPRC at any prevalence. Brier score was higher with a correction in 11 of 12 cells at 1:20 and 1:50 but without statistical support, and the absolute calibration intercept was not consistently larger. What the corrections did for LR was shift predicted risk upward: the signed intercept fell in every cell and went from underestimation to overestimation at 1:5–1:50. At the 0.5 cut-off this raised LR sensitivity at every prevalence and balanced accuracy at 1:5–1:50; threshold moving, which cannot change predicted risks, raised both more at 1:20 and 1:50. For GB neither helped reliably at 1:20 and 1:50, and the analytic prior-shift offset did not repair the corrected models. With one easy dataset and 10 seeds the advice is modest: report the signed calibration intercept with discrimination, and try the threshold before resampling when only the operating point matters.

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